



Mixed rumen bacteria grow with similar efficiency on cellulose and glucose

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ABSTRACT

Microbes growing in the rumen represent over half of the protein digested by cattle. Despite this importance, efficiency of microbial growth can vary widely, and the reasons are not fully understood. Here, we investigated if the carbohydrate source (cellulose or glucose) affects growth efficiency. We inoculated a system of 8 fermentors with rumen fluid, fed them glucose or cellulose (30 mmol hexose L⁻¹·d⁻¹), and maintained them at a range of dilution rates (2% to 12%·h⁻¹). We then measured the digestion of carbohydrates and output of microbes and fermentation products. We found that distinct communities of bacteria grew in fermentors fed glucose versus cellulose, with *Streptococcus* and *Kandleria* in the former and *Fibrobacter*, *Ruminococcus*, and other groups in the latter. Additionally, they formed different fermentation products, with cellulose-fed fermentors producing acetate and propionate and glucose-fed ones also producing butyrate and caproate. A slightly larger mass of microbes grew in fermentors fed glucose versus cellulose, but they also fermented more carbohydrates. As a result, microbes in glucose- and cellulose-fed fermentors grew with similar efficiency. These results suggest that the carbohydrate source may not be an important factor in determining efficiency of microbial growth in the rumen, but it does alter the composition of microbial communities and fermentation products.

Key words: rumen, bacteria, growth, cellulose, glucose

INTRODUCTION

Cattle are unique because most of the protein available for digestion does not come directly from the diet, but rather from microbes growing in the rumen. Across studies, 68% of crude protein flowing from the rumen

has been found to be microbial in origin (Broderick et al., 2010). These microbes grow by fermenting dietary carbohydrates, and their protein-rich cells then pass to the small intestine for digestion. In addition to being abundant, microbial protein is inexpensive (Satter et al., 2002); microbes can synthesize it from urea or NPN, which are less expensive than true protein. Increasing microbial growth further would thus improve the protein supply for cattle and benefit the livestock industry.

Despite the importance of microbes growing in the rumen, efficiency of that growth varies widely (Hackmann and Firkins, 2015). Within and across experiments, reported efficiencies range approximately 2-fold. This variation is observed both in the rumen (Broderick et al., 2010) and in laboratory cultures of rumen microbes (Russell and Wallace, 1997). One factor known to affect efficiency is the amount of carbohydrate provided to cells (Hackmann and Firkins, 2015). Providing excess carbohydrates depresses efficiency, likely due to cells synthesizing glycogen (Hackmann et al., 2013; Denton et al., 2015; Teixeira et al., 2017) or spilling energy as heat (Russell, 2007; Hackmann et al., 2013; Teixeira et al., 2017). This factor may explain why microbes grow inefficiently in cows fed large amounts of grain (Clark et al., 1992). However, it does not explain low efficiency in other circumstances.

The type of carbohydrate, not just the amount, should be explored for its potential effects on growth efficiency. Microbes from the rumen ferment a wide range of carbohydrates, with cellulose and other carbohydrates of the plant cell wall being the most abundant. Sugars such as glucose are also present in the rumen, but at far lower concentrations (Kajikawa et al., 1997). Because they are soluble and easy to feed, sugars are almost always used to measure growth efficiency in microbes in laboratory cultures (Russell and Wallace, 1997). In the few cases that cellulose has been fed (Weimer et al., 1991; Shi and Weimer, 1992; Weimer, 1993), the growth efficiency has not been directly compared with that of cultures fed sugars. Comparing different experiments (Weimer, 1993; Wells and Russell, 1994) suggests that

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

one bacterium (*Fibrobacter succinogenes* S85) grew more efficiently on cellulose than cellobiose (a disaccharide). However, this comparison is not direct, and it is necessary to measure efficiency on different carbohydrates within the same experiment.

Our objective was to measure efficiency of mixed bacteria growing in fermentors fed glucose versus cellulose. We used fermentors to precisely control conditions such as dilution rate, and we focused on bacteria because they make up between 60% and 90% of the rumen microbial biomass (Lin et al., 1997; Hackmann, 2025a). Our findings show that under our conditions, the source of carbohydrate (glucose or cellulose) did not affect growth efficiency. This work will help to narrow down the search for other factors that affect efficiency, which can help formulate better rations for cattle.

MATERIALS AND METHODS

Construction of Fermentor System

Our experiments used a system of 8 fermentors. The system is based on a design described in Wenner et al. (2021) and shown in detail in Figure 1 and Supplemental Figure S1 (see Notes). In brief, it consists of glass jars (The Ohio State University) that hold the liquid contents. Each jar has a magnetically driven impeller to mix the contents, and the contents are kept at 39°C with a heating rod and temperature probe. An overflow port on the jar allows contents to flow out with the input of buffer. With the overflow port, the volume of liquid in the jars is maintained at 1.85 L. A pressure relief valve allows gas from fermentation to be vented, and the system operates under positive pressure. The jar sits on a base unit (The Ohio State University) that provides electronic control over stirring and heating. Peristaltic pumps (Cole-Parmer 07528-30 and 77202-60) were used to infuse buffers from tanks. Unlike in Wenner et al. (2021), pumps were not used to remove liquid contents from jars. The tanks of buffer were bubbled with CO₂ (anaerobic grade; Airgas CD AN200) at ~100 mL·min⁻¹ and mixed with an impeller. Bubbling the tank alone was adequate in maintaining low redox (Supplemental Figure S8, see Notes). Effluent collection jars (5-L Pyrex bottles; Corning) were outfitted to hold fermentor effluent, connected to the jars with large tubing, and held at 0°C in a large ice bath.

Feeding glucose and cellulose was accomplished with devices described in Figure 1. Glucose was delivered as a solution using a simple pump (Figure 1B). The peristaltic pumps for the glucose-feeding device were the same as those used for infusing buffers. For the cellulose-feeding device, the syringe pump was a KD Scientific 780100, the magnet for the stirrer was a NI B888-N52 (K&J Magnetics), and the disk magnet for the syringe was a K&J

Magnetics DE2. The magnet in the syringe was stirred at 30 rpm, which was measured using a tachometer. Because cellulose consists of particles that are insoluble in water, feeding it requires the design of a specialized device (Figure 1C). This device consisted of a syringe filled with cellulose slurry. This slurry was mixed magnetically and delivered into the fermentors with a syringe pump.

The accuracy of the cellulose-feeding device was measured in 2 ways. In preliminary experiments, we measured delivery of cellulose in aluminum dishes over the course of the day. These experiments showed that cellulose is delivered at 101.23% (SEM 0.93%) of the expected delivery rate (0.4 g·h⁻¹). During the main experiment, the actual amount of cellulose delivered was measured as described in the following section. The actual amount was 108.2% (SEM 3.4%) of the target amount (Supplemental Figure S2, see Notes).

Initial Operation of Fermentors

On the first day of each trial, rumen fluid was collected from 3 ruminally-cannulated heifers. Heifers were fed a diet consisting on a DM basis of 50.6% wheat hay, 24.6% alfalfa hay, 21.1% almond hulls, and 3.7% UC Davis Dry Cow Pellet (Western Milling, LLC). As determined by wet chemistry methods (Cumberland Valley Analytical Services), chemical composition was 91.8% OM, 43.8% NDF analyzed with heat-stable amylase and expressed exclusive of residual ash (aNDFom), 31.2% ADF, 12.7% CP, and 6.4% starch (DM basis). The procedure for measuring NDF followed Van Soest et al. (1991), except filtering was done with a Whatman 934-AH glass microfiber filter with 1.5-μm particle retention, and the final glass fiber filter and sample were ashed in a 535°C furnace for 2 h. Fluid was collected with a siphoning device as illustrated in DePeters and George (2014). The time of collection was at 0800 h (~3 h after feeding).

Fermentors were inoculated with rumen fluid within 1 h of collection. Fermentor jars were prefilled with ~950 mL of warm, reduced buffer, then 900 mL of rumen fluid were added. Jars were flushed with CO₂, and the lids were then put back in place.

Feeding of glucose and cellulose was started immediately after inoculation. Glucose was delivered to a target amount of 10 g·d⁻¹ by infusing a solution (119 g·L⁻¹) from large bottles. Glucose solution was prepared daily and replaced every 12 h. Solutions were not sterilized, and a preliminary experiment showed no loss (microbial degradation) of glucose in the bottle over a period of 2 d. Cellulose (Sigma S3X504-500G) was delivered to a target amount of 8.9 g·d⁻¹ using the feeding device described previously. The nominal particle size of this cellulose is 20 μm. Syringes were loaded with slurry (200 g·L⁻¹) and a disk magnet, then placed in the syringe pump. Syringes

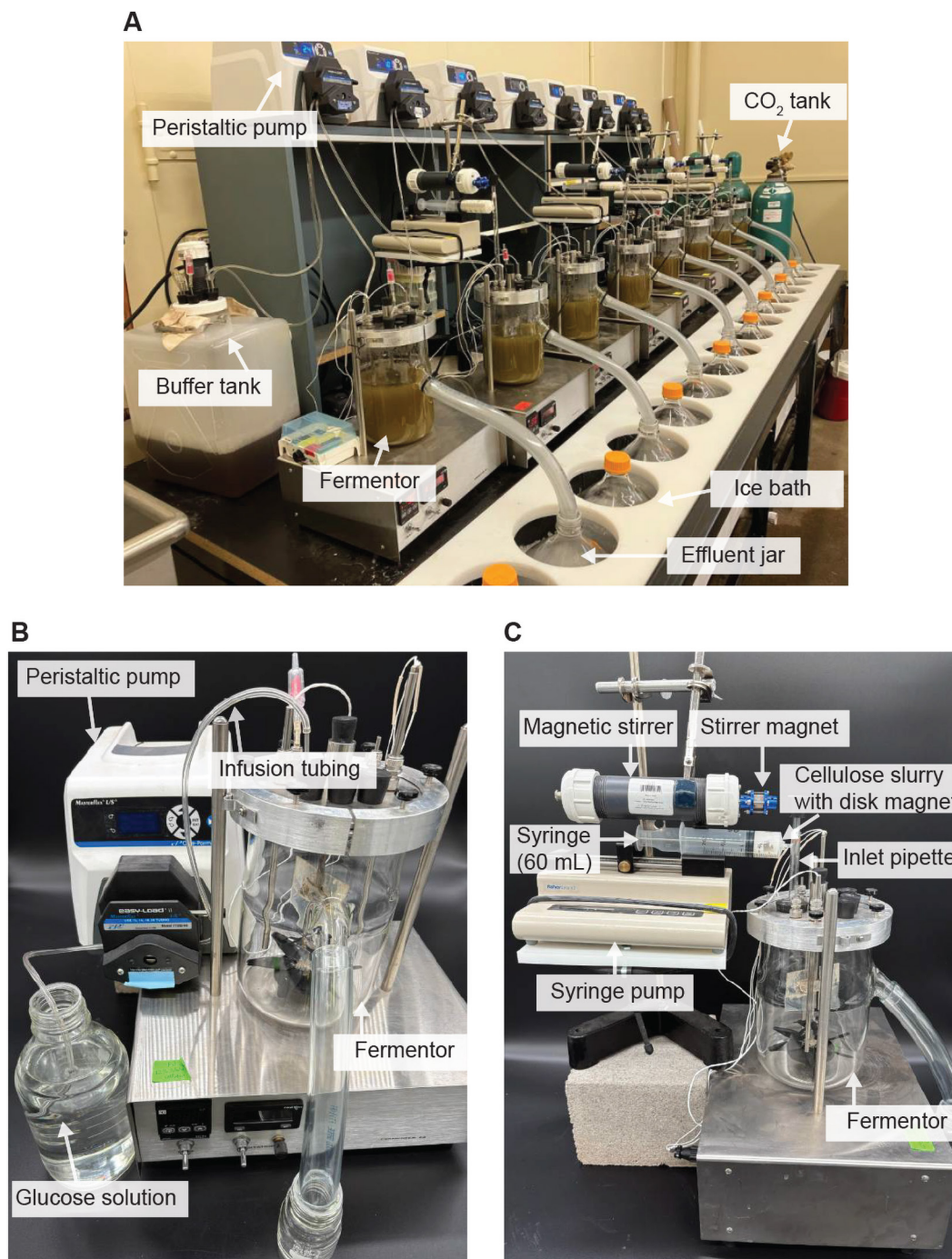


Figure 1. We constructed a system of 8 fermentors capable of continuously feeding glucose or cellulose. (A) Overview of fermentors. (B) Glucose-fed fermentor, showing the simple device for feeding glucose. (C) Cellulose-fed fermentor, showing syringe-based device for feeding cellulose.

were changed twice daily, leaving ~15 mL of slurry in the syringe before changing. To verify the amount delivered, the remaining slurry was collected, dried, and weighed.

The amount of carbohydrate fed was chosen to fall in the middle of the range of Isaacson et al. (1975), but it

was also partly determined with a preliminary experiment. In the preliminary experiment, a fermentor was fed double the normal amount of cellulose (total of 60 mmol hexose·L⁻¹·d⁻¹), and the pH was decreased to a lower value than desired (5.9). After resuming the normal rate

(30 mmol hexose $\cdot$ L⁻¹ $\cdot$ d⁻¹), the pH rose to a higher value (6.1). This showed the normal amount of cellulose was compatible with our system, but the high amount was not.

Buffer was infused continuously throughout the trial at target dilution rates of 2%, 4%, 8%, and 12% $\cdot$ h⁻¹. This corresponds to a volume of buffer of 0.037 to 0.222 L $\cdot$ h⁻¹. The composition of buffer was slightly modified from Isaacson et al. (1975). Per liter, the buffer included 0.4 g of Trypticase peptone (product 211921, Thermo Scientific), 0.1255 g of (NH₄)₂SO₄, 0.7177 g of urea, 0.9665 g of KH₂PO₄, 0.9732 g of K₂HPO₄, 0.9596 g of NaCl, 0.1294 g of CaCl₂ $\cdot$ H₂O, 0.0975 g of MgSO₄, 6.341 g of NaHCO₃, 10 mL of Pfennig's heavy metal solution, and 200 mL of clarified rumen fluid. Per liter, the Pfennig's heavy metal solution included 0.5 g of EDTA disodium salt, 0.2 g of FeSO₄ $\cdot$ 7H₂O, 0.2 g of MnCl₂ $\cdot$ 4H₂O, 0.01 g of ZnSO₄ $\cdot$ 7H₂O, 0.03 g of H₃BO₄, 0.02 g of CoCl₂ $\cdot$ 6H₂O, 0.001 g of CuCl₂ $\cdot$ 2H₂O, 0.002 g of NiCl₂ $\cdot$ 6H₂O, and 0.003 g of NaMoO₄ $\cdot$ 2H₂O (Schaefer et al., 1980). Resazurin (a redox indicator) was not included because we measured redox with a probe. To prepare clarified rumen fluid, rumen fluid was collected from ruminally-cannulated heifers as described previously. The fluid was centrifuged (10,000 $\times$ g, 10 min, 4°C; F14-6 $\times$ 250LE rotor and Sorvall Legend XTR centrifuge) to remove solid material, autoclaved at 121°C for 20 min, and stored at -20°C. Rumen fluid was thawed and buffer prepared immediately before use. The pH of the buffer (6.7 to 6.8) was measured to ensure consistency between batches.

Fermentors were run for 7 d to enable adaptation to the carbohydrate and other conditions. This 7-d period allowed contents of fermentors to turn over a minimum of 3.4 times. Early experiments showed that pH and redox had stabilized within this period (Supplemental Figure S3, see Notes). Sequencing of the 16S rRNA gene of bacteria also showed that the communities had started to adapt by the end of 1 d and did not change between 8 and 12 d (Supplemental Figure S3). This is similar to other work showing bacterial communities in fermentors stabilize within 4 d (Salfer et al., 2018).

Feeding of glucose and cellulose was done continuously to provide constant (steady-state) conditions in the fermentors. In a preliminary experiment, we verified that fermentors had stable pH over the day (Supplemental Figure S4, see Notes).

The experiment followed a completely randomized design, with treatments (carbohydrate and dilution rate) being assigned randomly to fermentors. This randomization controlled any confounders (e.g., location of individual fermentors). All authors were aware of assignment of treatments. Experiments were repeated 3 times (for a total of 3 trials). The start dates of the trials were April 17, 2022, March 13, 2023, and May 17, 2023.

Measurement and Sampling of Fermentors

After adaptation, we began formal sampling and measurement of fermentors. This sampling and measurement period lasted from d 8 to d 12. At ~0800 h and 1800 h, effluent collected in jars was weighed to determine the total outflow of effluent (g $\cdot$ d⁻¹). At the 0800 h time point, small samples of effluent (three 1-mL samples and one 50-mL sample) were collected in centrifuge tubes. Additionally, a large sample (1 L) was collected in a plastic container. The 1-mL samples were collected for fermentation product assays (VFA, ammonia, lactic acid, and glucose), the 50-mL samples were collected for cell pellet DM and bacterial sequencing, and the 1-L samples were collected for effluent DM and NDF analysis. The small samples were held on ice until processing, and the large sample was stored at -20°C. The remaining effluent was discarded after each time point (0800 h and 1800 h).

Contents from the fermentor jars and buffer tank were sampled similarly. At 0800 h of each day, we collected two 2-mL samples directly from each of the fermentor jars and the buffer tank for fermentation product assays. Unlike the effluent, where 1- to 2-mL, 50-mL, and 1-L samples were taken, only the 1- to 2-mL samples were collected from the fermentor jars and buffer tank during daily sampling. On d 12, an additional 250-mL sample was taken from each fermentor jar. At the same time, pH and redox potential were measured by placing probes in the jars and tank.

Samples of effluent, buffer, and fermentor contents were held on ice and processed within 2 h of collection. To harvest bacterial cells, samples were centrifuged (10,000 $\times$ g, 10 min, 4°C). The cell pellet was washed twice in NaCl (0.9% wt/vol) and stored at -20°C for analysis. For the 2-mL fermentor samples, supernatant from the first centrifugation was kept and stored at -20°C for analysis. For the 1-mL effluent and 2-mL buffer samples, centrifugation was done using a 24 $\times$ 1.5/2.0-mL rotor and Sorvall Legend 21R centrifuge. For 50-mL effluent samples, we used a F15-8 $\times$ 50cy rotor and Sorvall Legend XTR centrifuge. For 250-mL fermentor samples, we used a F14-6 $\times$ 250LE rotor and Sorvall Legend XTR centrifuge.

After the trial, large samples of effluent were thawed, transferred to a glass dish, and air-dried in an oven for 48 h at 55°C. The dried material was weighed, homogenized in a coffee grinder (Hamilton Beach 80335R), then stored at room temperature until further analysis.

Collection of Other Samples

Cell pellets were prepared from the rumen fluid of cows. Rumen fluid was collected from one of 3 heifers. A 200-mL volume was centrifuged at a low force (1,000 $\times$ g, 5 min, 4°C; F15-8 $\times$ 50cy rotor and Sorvall Legend XTR

centrifuge) to remove large feed particles and protozoa. A 50-mL aliquot of supernatant was centrifuged at high force ($10,000 \times g$, 10 min, 4°C) to harvest bacterial cells. The pellet was washed twice with NaCl (0.9% wt/vol), then stored at -20°C for further analysis. This experiment was repeated on 3 d.

To supplement 2-mL samples of buffer collected during the main trial, we also collected 250-mL aliquots of buffer in a separate experiment. We air-dried these samples following the same procedure used for the effluent.

Analysis of Fermentor Samples

Samples of air-dried effluent were analyzed for DM. This was determined by drying ~ 0.5 g of material in aluminum dishes (VWR 25433-008) at 105°C .

Samples of air-dried effluent were also analyzed for NDF. This involved a modification of the micro-NDF procedure described previously (Weimer et al., 1990; Pell and Schofield, 1993). To a 125-mL serum bottle, we added 0.1 g of sample, 20 mL of neutral detergent solution (ANKOM FND20), 0.2 g of Na_2SO_3 , and 40 μL of heat-stable α -amylase (19,683 Liquefon Units $\cdot\text{g}^{-1}$; ANKOM FAA). The bottle was stoppered, capped, incubated at room temperature for 3 h, and then autoclaved at 121°C for 60 min. Bottles were held at 105°C , then contents were poured through a filter (GF/D, 47 mm; Cytiva 1823047) on a Buchner funnel. The bottle and filter were washed with 3 volumes of 50 mL of hot (90°C – 100°C) water, 50 mL of room-temperature acetone, and 50 mL of room-temperature ethanol. The filter was dried for 24 h at 105°C to determine the mass of NDF. To correct for any loss of filter during washing, we processed blank samples (bottles containing no sample but neutral detergent solution and all other components). When analyzed with this method, cellulose was found to be 95.2% (SEM 5.5%) NDF ($n = 5$). For comparison, analysis was also done by a commercial laboratory (Cumberland Valley Analytical) using the standard NDF method (Van Soest et al., 1991), and cellulose was found to be 95.1% (SEM 1.7%; $n = 4$) NDF. This shows that our method produces the same results as the standard method, though with higher variability. Additionally, similar values were found for cellulose slurry after being mixed in syringes for 24 h.

Samples of fermentor contents were analyzed for fermentation products, lactate, ammonia, and glucose. Analysis of most fermentation products (acetate, propionate, butyrate, isobutyrate, isovalerate, caproate) was done with gas chromatography using the conditions of Zhang et al. (2021). The values reported for analysis cannot distinguish isovalerate and 2-methylbutyrate, which co-elute under the conditions of our analysis. Analysis of lactate was done with an enzymatic kit (Zhang et al., 2021), and the values reported are the sum of D- and L- isomers. Glu-

cose was analyzed using the glucose oxidase-peroxidase method (Zhang et al., 2021). Ammonia was analyzed using phenol-hypochlorite (Broderick and Kang, 1980).

Samples of cell pellets were analyzed for DM, NDF, N, C, and H. For DM, we analyzed cell pellets from 50-mL aliquots of effluent, 200-mL aliquots of rumen fluid, and 250-mL aliquots of fermentor contents. Analysis of DM was done by drying material in aluminum dishes as previously described. For NDF, N, C, and H, we analyzed cell pellets from 250-mL aliquots of fermentor contents. Analysis of NDF was done using the micro-NDF procedure as previously described. Analysis of N, C, and H was done using an elemental analyzer (Perkin Elmer PE2400 Series II) by a commercial service (NuMega Resonance Labs).

Samples of buffer were analyzed for DM and NDF. In the case of NDF analysis, the value was -0.43% (SEM 0.40%) NDF, which was not different from 0 ($P = 0.3903$) and not considered in further calculations.

Calculation of Outflow and Other Values

We calculated outflow of bacteria, cellulose, DM, and fermentation products from fermentors. The outflow of bacterial DM ($\text{g}\cdot\text{d}^{-1}$) was calculated as

$$\begin{aligned} &\text{Bacterial DM in effluent (g}\cdot\text{g}^{-1}) \\ &\times \text{outflow of total DM (g}\cdot\text{d}^{-1}). \end{aligned}$$

The outflow of total DM ($\text{g}\cdot\text{d}^{-1}$) was

$$\text{DM in effluent (g}\cdot\text{g}^{-1}) \times \text{total outflow of effluent (g}\cdot\text{d}^{-1}).$$

The DM in effluent ($\text{g}\cdot\text{g}^{-1}$) was calculated from the amount of air-dried effluent and the DM of that material. Bacterial DM in effluent ($\text{g}\cdot\text{g}^{-1}$) was calculated as

$$\begin{aligned} &[\text{DM of cell pellet from effluent (g)} \\ &- \text{NDF of cell pellet from effluent (g)}] \\ &\times [1 + \text{NDF of cell pellet from rumen fluid (g}\cdot\text{g}^{-1})]. \end{aligned}$$

This calculation involves subtracting off the NDF of the cell pellet (g), which corrects for cellulose pelleted with cells during centrifugation. It also involves multiplying by the NDF of cell pellet from rumen fluid ($\text{g}\cdot\text{g}^{-1}$), which accounts for NDF in cells (0.042 [0.59 SEM] g NDF g DM^{-1}). The calculation assumes that buffer contributes no DM to the cell pellet. This was also assumed by Isaacson et al. (1975), and we verified it under our conditions by pelleting and analyzing samples of buffer. We found these samples had 0.0050 (SEM 0.0503) g $\text{DM}\cdot\text{L}^{-1}$, which is not different ($P = 0.9229$) from 0.

The outflow of bacterial N ($\text{g}\cdot\text{d}^{-1}$) was calculated as

$$\begin{aligned} & \text{Outflow of bacterial DM (g}\cdot\text{d}^{-1}) \\ & \times \text{N of cell pellet from fermentor (g}\cdot\text{g}^{-1}). \end{aligned}$$

The calculation assumes that buffer contributes no N to the cell pellet.

For outflow of cellulose ($\text{g}\cdot\text{d}^{-1}$), the calculation was

$$\begin{aligned} & \text{Outflow of NDF (g}\cdot\text{d}^{-1}) \\ & - \text{outflow of bacterial NDF (g}\cdot\text{d}^{-1}). \end{aligned}$$

The outflow of NDF ($\text{g}\cdot\text{d}^{-1}$) was

$$\begin{aligned} & \text{NDF in effluent (g}\cdot\text{g}^{-1}) \\ & \times \text{total outflow of effluent (g}\cdot\text{d}^{-1}). \end{aligned}$$

The outflow of bacterial NDF ($\text{g}\cdot\text{d}^{-1}$) was

$$\begin{aligned} & \text{NDF of cell pellet from rumen fluid (g}\cdot\text{g}^{-1}) \\ & \times \text{outflow of bacteria (g}\cdot\text{d}^{-1}). \end{aligned}$$

For outflow of fermentation products ($\text{mol}\cdot\text{d}^{-1}$), the calculation was

$$\begin{aligned} & [\text{fermentation product in fermentor (mol}\cdot\text{L}^{-1}) \\ & \times \text{total outflow of effluent (g}\cdot\text{d}^{-1})/1,000 \text{ g}\cdot\text{L}^{-1}] \\ & - [\text{fermentation product in buffer (mol}\cdot\text{L}^{-1}) \\ & \times \text{total outflow of effluent (g}\cdot\text{d}^{-1})/1,000 \text{ g}\cdot\text{L}^{-1}]. \end{aligned}$$

This calculation assumes that fermentation products were in the same concentration in the effluent as in the fermentor contents. It also assumes that the flow of buffer was the same as the outflow of effluent.

After calculating outflows, we calculated digestibility. The digestibility of cellulose was

$$\begin{aligned} & [\text{Cellulose digested (g}\cdot\text{d}^{-1}) \\ & / \text{input of cellulose (g}\cdot\text{d}^{-1})] \times 100\%, \end{aligned}$$

where cellulose digested ($\text{g}\cdot\text{d}^{-1}$) was

$$\text{Input of cellulose (g}\cdot\text{d}^{-1}) - \text{outflow of cellulose (g}\cdot\text{d}^{-1}),$$

and the input of cellulose was that delivered by our feeding device. The digestibility of glucose was calculated similarly. The digestibility of DM was

$$[\text{DM digested (g}\cdot\text{d}^{-1})/\text{input of DM (g}\cdot\text{d}^{-1})] \times 100\%,$$

where DM digested ($\text{g}\cdot\text{g}^{-1}$) was

$$\begin{aligned} & \text{Input of DM (g}\cdot\text{d}^{-1}) - [\text{outflow of DM (g}\cdot\text{d}^{-1}) \\ & - \text{outflow of bacteria (g}\cdot\text{d}^{-1})], \end{aligned}$$

and the input of DM was the sum of DM from glucose, cellulose, and buffer.

Microbial efficiency was calculated in 4 ways. The first calculation was

$$\begin{aligned} & \text{Outflow of bacterial DM (g}\cdot\text{d}^{-1}) \\ & / [\text{glucose fermented (mol}\cdot\text{d}^{-1}) + \\ & \text{Cellulose fermented (mol}\cdot\text{d}^{-1})], \end{aligned}$$

where glucose fermented is

$$\text{Glucose digested (g}\cdot\text{d}^{-1})/180.156 \text{ g}\cdot\text{mol}^{-1} \times 0.92,$$

and cellulose fermented is

$$\text{Cellulose digested (g}\cdot\text{d}^{-1})/162.14 \text{ g}\cdot\text{mol}^{-1} \times 0.92.$$

The value of 0.92 is the fraction of digested carbohydrate that is fermented, whereas the remainder was assumed as microbial biomass. This value was measured using ^{14}C -glucose as in Isaacson et al. (1975), and we assume it is similar with cellulose.

The second calculation was as follows:

$$\begin{aligned} & \text{outflow of bacterial DM (g}\cdot\text{d}^{-1}) \\ & / \text{formation of ATP (mol}\cdot\text{d}^{-1}), \end{aligned}$$

where formation of ATP ($\text{mol}\cdot\text{d}^{-1}$) is

$$\begin{aligned} & \text{Outflow of acetate (mol}\cdot\text{d}^{-1}) \times 4 \\ & + \text{outflow of propionate (mol}\cdot\text{d}^{-1}) \times 3.5 \\ & + \text{outflow of isobutyrate (mol}\cdot\text{d}^{-1}) \times 4.5 \\ & + \text{outflow of butyrate (mol}\cdot\text{d}^{-1}) \times 4.5 \\ & + \text{outflow of isovalerate (mol}\cdot\text{d}^{-1}) \times 3 \\ & + \text{outflow of valerate (mol}\cdot\text{d}^{-1}) \times 3 \\ & + \text{outflow of caproate (mol}\cdot\text{d}^{-1}) \times 4.5 \\ & + \text{outflow of lactate (mol}\cdot\text{d}^{-1}) \\ & \times 2 + \text{cellulose fermented (mol}\cdot\text{d}^{-1}) \times 0.5. \end{aligned}$$

The coefficients in the equation come from Supplemental Table S1 (see Notes), which are from previ-

ous work (Hackmann et al., 2013; Wang et al., 2022; Hackmann, 2024).

The dilution rate was calculated as follows:

$$\frac{\text{Total outflow of effluent (g}\cdot\text{d}^{-1})}{1.85 \text{ L}/1,000 \text{ g}\cdot\text{L}^{-1}/24 \text{ h}\cdot\text{d}^{-1}} \times 100\%,$$

where 1.85 L refers to the working volume of the fermentor jar.

Methane was calculated from other fermentation products as follows:

$$\begin{aligned} & \text{Outflow of acetate (mol}\cdot\text{d}^{-1}) \times 1 \\ & + \text{outflow of propionate (mol}\cdot\text{d}^{-1}) \times -0.5 \\ & + \text{outflow of isobutyrate (mol}\cdot\text{d}^{-1}) \times -0.5 \\ & + \text{outflow of butyrate (mol}\cdot\text{d}^{-1}) \times 0.5 \\ & + \text{outflow of isovalerate (mol}\cdot\text{d}^{-1}) \times -0.25 \\ & + \text{outflow of valerate (mol}\cdot\text{d}^{-1}) \times -0.25 \\ & + \text{outflow of caproate (mol}\cdot\text{d}^{-1}) \times 0.5. \end{aligned}$$

The coefficients in the equation come from Supplemental Table S1, assuming all H_2 formed was used for methane.

The third and fourth calculations for microbial efficiency are

$$\begin{aligned} & \frac{\text{Outflow of bacterial N (g}\cdot\text{d}^{-1})}{[\text{glucose fermented (mol}\cdot\text{d}^{-1}) \\ & + \text{cellulose fermented (mol}\cdot\text{d}^{-1})]} \end{aligned}$$

and

$$\frac{\text{Outflow of bacterial N (g}\cdot\text{d}^{-1})}{\text{formation of ATP (mol}\cdot\text{d}^{-1})}.$$

These are similar to the first 2 calculations, but with outflow of bacterial N in the numerator.

Imaging and Sequencing of Bacteria

Fermentor contents were imaged by microscopy using an Axio Scope.A1 microscope (Zeiss, Munich, Germany), 100 $\times$ oil (A-Plan; numerical aperture, 1.25) objective, and an Axiocam 305 mono camera. Samples were prepared as wet mounts on a thin layer of 2% Noble agar. Zeiss ZEN version 3.9 was used for image acquisition and to linearly adjust brightness and contrast. The adjustment was applied to whole images.

The 16S rRNA gene of bacteria was sequenced following the methods in Tao et al. (2019). Extraction of DNA was done with the DNeasy PowerSoil kit (Qiagen), then the V4 region of the 16S rRNA gene was PCR-amplified with primers 515FB and 806RB (Thompson et al., 2017). Amplicons were purified with AMPure XP bead kit (Beckman Coulter) then sequenced on the Illumina MiSeq platform, generating 250-bp paired-end reads. These procedures were completed at Molecular Research LP.

After demultiplexing, sequence reads were clustered into amplicon sequence variants (ASV) with R (v. 4.4.1; R Core Team, 2024) and the DADA2 package (v. 3.20; Callahan et al., 2016). Taxonomic assignment of ASV was done at the genus level using a naive Bayesian classifier (Wang et al., 2007) and a taxonomic reference database. We used a reference database of 20,278 type strains from the List of Prokaryotic names with Standing in Nomenclature (LPSN; Meier-Kolthoff et al., 2022). We downloaded taxonomy and 16S rDNA sequences for these strains according to Hackmann (2025b). A second reference database (SILVA v. 138.2; Quast et al., 2013) was used in early work, but it had outdated names for key taxa (e.g., *Christensenellaceae R-7 group* instead of *Aristaella*; Supplemental Figure S5, see Notes). Analyses of α and β diversity were done using R and the phyloseq package (McMurdie and Holmes, 2013).

Samples for sequencing were cell pellets from fermentor contents. Samples collected on 1, 8, and 12 d were those analyzed. Metadata on samples and ASV are given in Supplemental Datasets S1 and S2 (see Notes).

Calculation of Digestion Rate of Cellulose

We calculated digestion rate of cellulose as follows:

$$\frac{\text{Digestibility (\%)} \times \text{dilution rate (\%}\cdot\text{h}^{-1})}{[100 - \text{digestibility (\%)}]}.$$

This equation assumes that digestion followed first order kinetics and all cellulose was potentially digestible.

Prediction of Traits with Fermentation Explorer

We predicted traits of ASV with Fermentation Explorer (v. 1.15; <https://github.com/thackmann/FermentationExplorer/>). Specifically, we used the “Predict from taxonomy” tool and taxonomy (phylum to genus) as an input query. As described in Hackmann and Zhang (2023), the tool makes predictions by matching the query against organisms in the internal database. If the matching organisms share a trait, the trait is predicted for the query. All predicted traits are in Supplemental Dataset S2.

Since its original description, Fermentation Explorer has received the following updates: The total number of organisms in the database is now 20,806. Names and taxonomy of organisms are now from the LPSN database and obtained as described in Hackmann (2025b). Phenotypic information now also comes from the *Anaerobe Laboratory Manual* (Holdeman et al., 1977; for $n = 192$ organisms). Information on metabolism of methanogens ($n = 131$) is now included. In total, fermentation or methanogenesis are reported for 2,581 organisms, and metabolic end products are reported for 1,600 organisms. The “Predict traits from taxonomy” tool was updated to predict all traits in the database (not just those related to fermentation). Other updates are described on Fermentation Explorer’s GitHub page (<https://github.com/thackmann/FermentationExplorer>).

Statistical Analysis

Data were read into R and screened for outliers. We removed all data from 2 fermentor-day-trial combinations due to unusually low values of NDF digestibility, and we removed data from 1 fermentor-day-trial combination due to unusually low VFA outflow values. This left 117 fermentor-day-trial combinations (98% of the total) for analysis. Unless otherwise stated, data were averaged across days within trial.

We fit values of efficiency to the model of Pirt (1965) using the nlme package and nlme() function in R (Pinheiro et al., 2023). The model used was as follows:

$$\text{Efficiency} \sim 1/(\text{Maintenance}/\text{Dilution} + 1/\text{Max_efficiency}).$$

Type of carbohydrate was a fixed effect, specified with the following argument:

$$\text{fixed} = \text{list}(\text{Maintenance} \sim \text{Carbohydrate}, \text{Max_efficiency} \sim \text{Carbohydrate}).$$

Fermentor was a random effect, specified with the following argument:

$$\text{random} = \text{Maintenance} + \text{Max_efficiency} \sim 1 \mid \text{Fermentor}.$$

Data were weighted using the varPower() function in R. Including trial as a random effect led to nonconvergence, so it was omitted from the final model. Standard errors were estimated with the emmeans package (Lenth, 2025) and emmeans() function. *P*-values for parameters are from nlme(). Confidence intervals (95%) were estimated

using bootstrapping with the nlraa package (Miguez, 2023) and predict_nlme() function.

The analysis above uses nonlinear regression, whereas Pirt (1965) used double-reciprocal plots and linear regression. To determine if regression method would affect our results, we also analyzed data with linear regression by using the lme() function and the following model:

$$\text{Inverse efficiency} \sim \text{Inverse dilution} \times \text{Carbohydrate}.$$

Fermentor was a random effect, specified with the following argument:

$$\text{random} = \sim \text{Inverse dilution} \mid \text{Fermentor}.$$

Maintenance is equal to the fitted slope, and maximum efficiency is equal to the inverse of the fitted intercept. Standard errors for maintenance were estimated using the emmeans package and emtrends() function, and SE for maximum efficiency were estimated using the emmeans() function and the delta method.

Differences in α diversity were determined using a Kruskal–Wallis test. When that test was significant ($P < 0.05$), means were separated using a Dunn test. We used the stats (R Core Team, 2024) and FSA packages (Ogle et al., 2025) and kruskal.test() and dunnTest() functions in R. Data were analyzed directly and not averaged across days within trial.

Difference in β diversity determined using permutational multivariate ANOVA. We used the vegan (Oksanen, 2025) and pairwiseAdonis packages (Martinez Arbizu, 2020) and adonis2() and pairwise.adonis() functions in R. When analyzing the effect of dilution rate, the actual dilution rate was rounded to the closest target rate (2%, 4%, 8%, or 12%·h⁻¹). Data were analyzed directly and not averaged across days within trial.

For plots of variables against dilution rate, regression was performed using the ggplot2 package (Wickham, 2016) and geom_smooth() function. The method chosen was gam, the basis was cubic splines, and the dimension was 4. Confidence intervals (95%) were those outputted by the geom_smooth() function. We constructed scatterplots using the ggplot2 package in R. Treemaps were constructed using the treemap package (Tennekes, 2023), with rectangles proportional to the relative abundance of ASV or their traits.

RESULTS

A Novel Feeding Device Allowed Continuous Feeding of Cellulose to Fermentors

To compare efficiency of microbial growth on cellulose versus glucose, we built a fermentor system that

can continuously feeding glucose or cellulose (Figure 1). Glucose was delivered as a solution using a simple pump (Figure 1B), whereas cellulose was delivered in a syringe full of slurry (Figure 1C). This slurry was mixed magnetically and delivered into fermentors with a syringe pump. We found this device resulted in stable delivery of cellulose over the day (see Materials and Methods).

With this feeding device in place, we established communities of bacteria growing on glucose or cellulose. We inoculated fermentors with mixed rumen contents, fed 30 mmol hexose equivalents $L^{-1} \cdot d^{-1}$, and infused buffer to maintain dilution rates between 2% and 12% $\cdot h^{-1}$. The actual input of glucose, cellulose, and buffer is shown in Supplemental Figure S2. After 7 d of adaptation, we sampled fermentor contents and their effluent (outflow) for analysis of bacteria and fermentation products. Although our ultimate aim was to determine the efficiency of bacterial growth, we started by characterizing the bacteria that grew and the fermentation they carried out.

Distinct Communities of Bacteria Grew in Glucose- and Cellulose-Fed Fermentors

After establishing fermentors, we began by characterizing the types of bacteria that grew in them. We found that distinct communities grew in glucose- and cellulose-fed fermentors, which was apparent with both microscopy (Figure 2A) and 16S rDNA sequencing (Figures 2B and 2C). Glucose-fed fermentors were predominated by chains of cocci and short rods (Figure 2A). Sequencing revealed these cells to be *Streptococcus* and *Kandleria* (Figure 2B). Cellulose-fed fermentors had a wider range of cell types (Figure 2A). Cocci and short rods were attached to cellulose particles, and the surrounding liquid had additional types of cells. Sequencing of these cells revealed *Aristaeella*, *Fibrobacter*, *Ruminococcus*, and unclassified groups (Figure 2B).

When we measured α and β diversity (Figure 2C), those data confirmed that glucose and cellulose-fed fermentors had distinct communities of bacteria. All measures of α diversity indicated a more diverse community in cellulose- versus glucose-fed fermentors. A nonmetric multidimensional scaling (NMDS) plot of Bray–Curtis distances, an expression of β diversity, also indicated these communities were different.

We found that communities were relatively stable across dilution rates. For glucose-fed fermentors, *Streptococcus* and *Kandleria* had steady abundance across dilution rates (Supplemental Figure S6A, see Notes). For cellulose-fed fermentors, *Aristaeella* and *Ruminococcus* were also stable, though *Fibrobacter* decreased to ~0 relative abundance at high dilution rates (Supplemental Figure S6A). Dilution rate also had few consistent effects on α and β diversity (Supplemental Figure S7, see Notes).

Fermentors also contained archaea. In total, they made up 1.90% (SEM 0.32%) of the total sequence reads for glucose-fed fermentors and 1.37% (SEM 0.41%) for cellulose-fed fermentors (see Supplemental Datasets S1 and S2). Nearly all were *Methanobrevibacter* (97.59% [SEM 0.74%] of total archaea), and they declined with increasing dilution rate (Supplemental Figure S6B). Protozoa were initially observed, but they had been lost by end of adaptation (d 7). Fungi were also not observed by the end of adaptation.

In sum, these results show that glucose- and cellulose-fed fermentors developed distinct bacterial communities from the same inoculum, and all communities also contained archaea. The formation of these distinct communities provides important insight for interpreting the associated fermentation products and growth efficiencies.

Bacteria Actively Fermented Glucose and Cellulose Fed to Fermentors

After characterizing the bacteria that grew in the fermentors, our next objective was to determine how they fermented the glucose and cellulose fed to them. When we measured fermentation products in fermentor contents, we found evidence of active fermentation in glucose- and cellulose-fed fermentors (Figure 3). For glucose-fed fermentors, the products in highest concentration were acetate and butyrate. At low dilution rates, caproate was also present. For cellulose-fed fermentors, acetate and propionate were highest. The total concentrations of all products decreased with increasing dilution rate, reflecting dilution with buffer and decreasing digestibility of cellulose. Methane was calculated from hydrogen balance, and it was produced under all conditions. We also observed values of pH, redox, and ammonia consistent with an active fermentation (Supplemental Figure S8, see Notes).

In addition to observing high concentrations of fermentation products, we found that glucose and cellulose were extensively digested within the fermentors. We calculated digestibility from the amount of glucose or cellulose fed and their respective outflow from the fermentors (see Materials and Methods). The digestibility of glucose approached 100% across dilution rates (Supplemental Figure S9, see Notes). The digestibility of cellulose ranged from a maximum of 92.1% (SEM 0.77%) at a dilution rate of 2% $\cdot h^{-1}$ to a minimum of 69.9% (SEM 5.3%) at a dilution rate of 12% $\cdot h^{-1}$ (Supplemental Figure S9). From these measurements, the digestion rate of pure cellulose is calculated to be 24.1% (SEM 2.7%) $\cdot h^{-1}$. Because digestion of glucose was essentially complete, its digestion rate was too high to define. Digestibility of DM, which includes buffer, was also high in both types of fermentors.

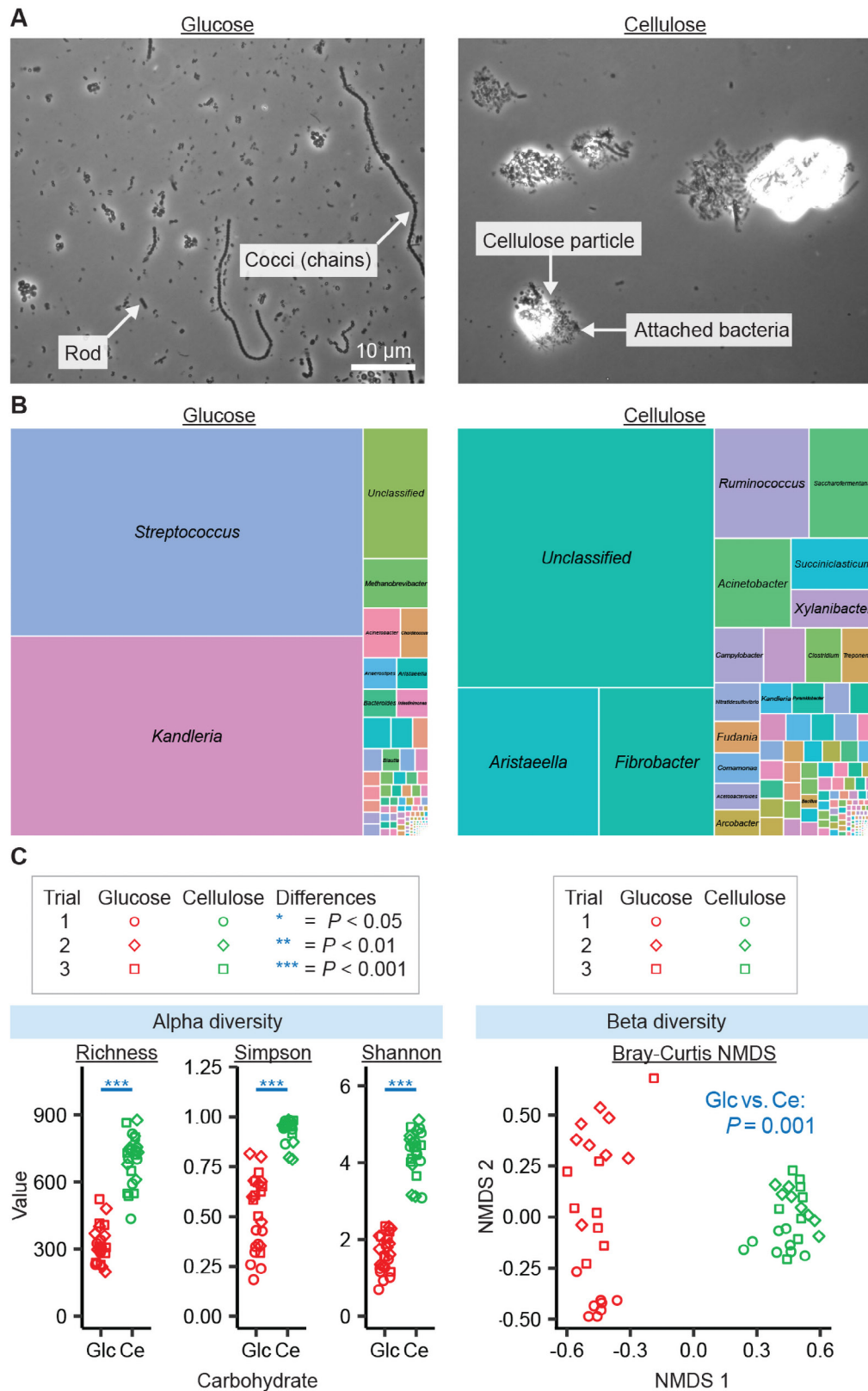


Figure 2. Distinct communities of bacteria grew in glucose- and cellulose-fed fermentors. (A) Morphology under microscope. (B) Genera of taxa identified with 16S rDNA sequencing. (C) Measures of α and β diversity of taxa identified with 16S rDNA sequencing. Communities also included archaea. Ce = cellulose; Glc = glucose; NMDS = nonmetric multidimensional scaling.

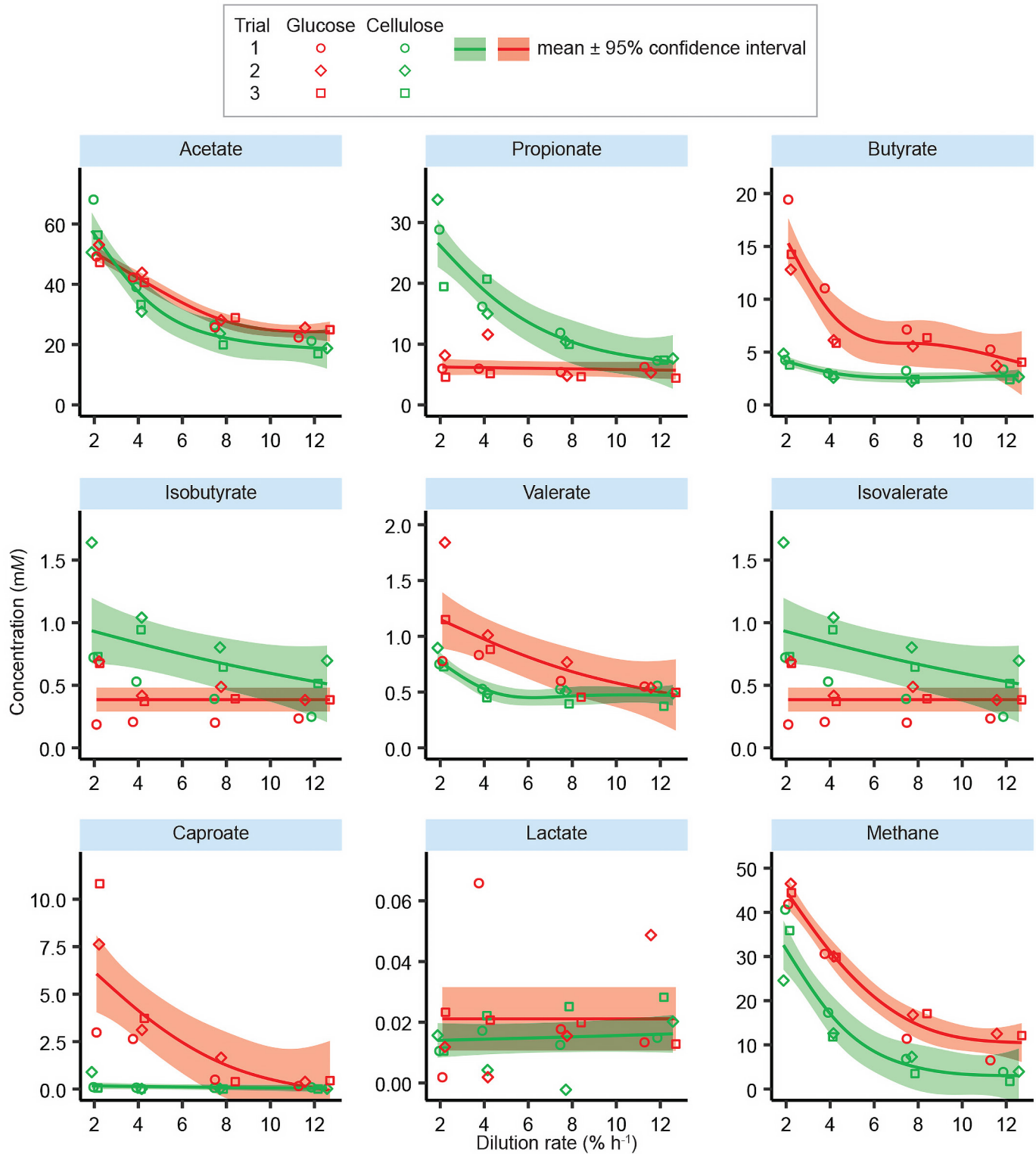


Figure 3. Bacteria fermented glucose and cellulose to a variety of end products. Concentrations refer to values in fermentor contents.

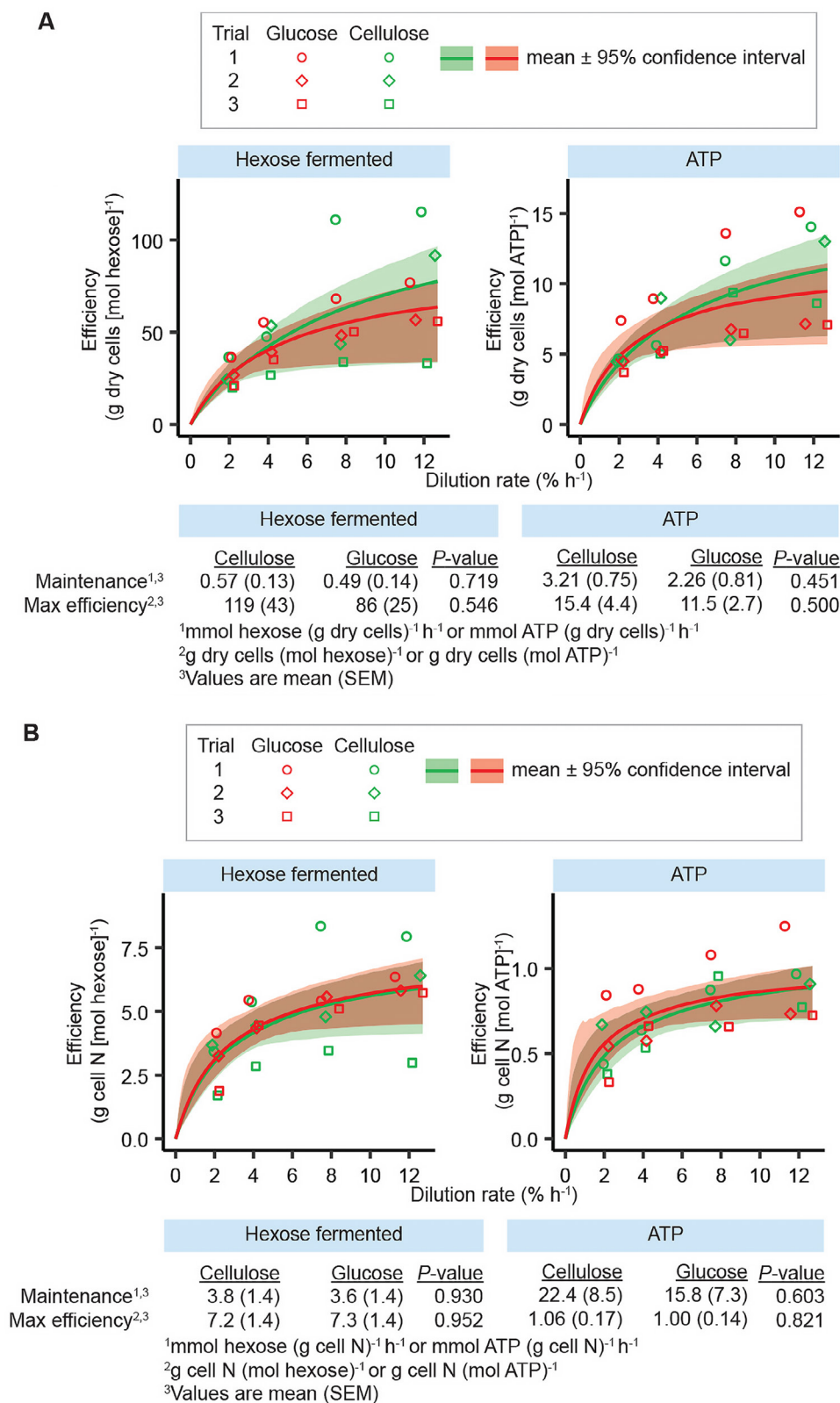


Figure 4. Bacteria grew with near equal efficiency on cellulose versus glucose. Efficiency is expressed in terms of (A) grams of dry cells (mol hexose)⁻¹ or grams of dry cells (mol ATP)⁻¹ or (B) grams of cell N (mol hexose)⁻¹ or grams of cell N (mol ATP)⁻¹. Shown is the relationship between dilution rate and efficiency and estimates of maintenance and maximum (max) efficiency. Fitted curve and estimates in are the result of fitting the model of Pirt (1965).

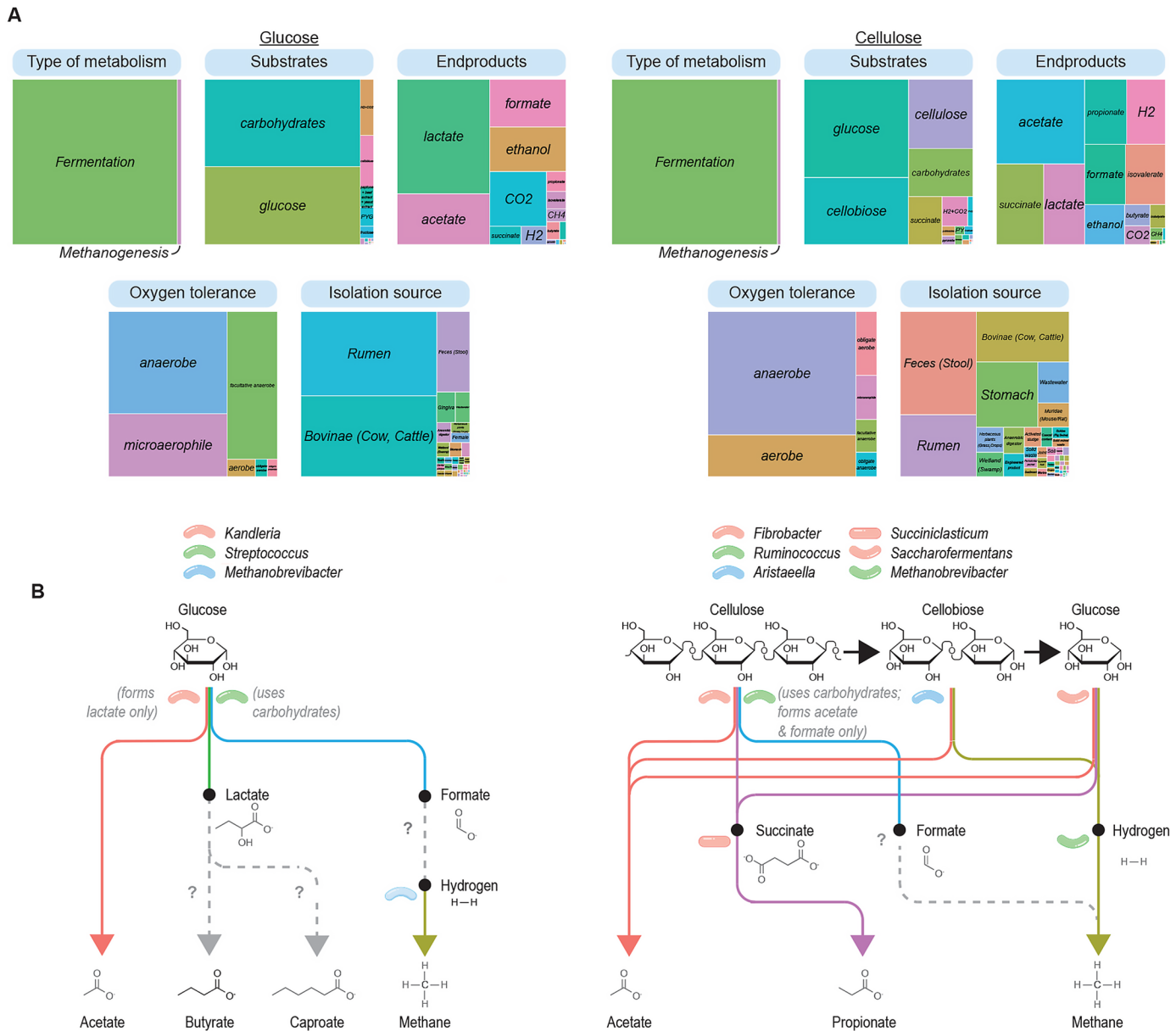


Figure 5. Fermentation Explorer–predicted traits of fermentation in fermentors. (A) Predicted traits. (B) Schematic of fermentation based on predictions.

These results show that bacteria carried out an active fermentation of glucose or cellulose. Concentrations of fermentation products were high, as were digestibilities of glucose and cellulose.

Bacterial Growth Was Higher on Glucose Versus Cellulose

After characterizing fermentation, we measured the amount of bacterial growth. We measured the outflow of fermentors in terms of DM ($\text{g DM} \cdot \text{d}^{-1}$) and nitrogen

($\text{g N} \cdot \text{d}^{-1}$). At low dilution rates, the outflow of bacteria was low and roughly equal in glucose- and cellulose-fed fermentors (Supplemental Figure S10, see Notes). The outflow increased with dilution rate, and it became slightly higher in glucose- than cellulose-fed fermentors. This difference was not significant when outflow was expressed in grams of DM per day (Supplemental Figure S10), but it was significant when expressed in grams of N per day. In summary, slightly more bacteria grew on glucose than cellulose, and this was most apparent at high dilution rates and when measuring nitrogen.

Bacteria Grew with Nearly Equal Efficiency on Cellulose Versus Glucose

With measurements of growth and other variables in hand, we were ready to calculate efficiency of growth (our main objective). We calculated efficiency using 4 different measurements (grams of dry cells per mole of hexose fermented, grams of cell DM per mole of ATP, grams of cell N per mole of hexose fermented, and grams of cell N per mole of ATP).

We found that regardless of measure, growth efficiency was similar across carbohydrates (Figure 4). For each carbohydrate, efficiency increased with dilution rate following a well-defined pattern. For some measures, efficiency was lower for cellulose at low dilution rates and higher at high dilution rates. However, this difference was not consistent and relatively small.

To analyze data in more detail, we fit them to a mathematical model that predicts efficiency from dilution rate and 2 parameters. The model was as follows:

$$\text{Efficiency} = 1/(\text{Maintenance coefficient} / \text{Dilution rate} + 1/\text{Maximum efficiency}).$$

The equation is from Pirt (1965), and units are defined in Figure 4. The first parameter, the maintenance coefficient, describes the amount of hexose (or ATP) expended by cells at zero growth rate. The second parameter, maximum efficiency, is the value of efficiency observed at infinite growth rate. We assumed dilution rate was equal to growth rate (the variable in Pirt, 1965). In Pirt (1965), efficiency is called the observed yield, and maximum efficiency is called the true growth yield.

We obtained estimated values of the maintenance coefficient and maximum growth efficiency after fitting the model to the data (Figure 4). The values of the maintenance coefficient were numerically higher for bacteria grown on cellulose versus glucose, but they had large SE and did not differ significantly. Values for maximum growth efficiency were also higher, but again not significantly and not across all measures of efficiency. We fit the model using nonlinear regression, and we obtained similar results when using linear regression instead (Supplemental Figure S11, see Notes).

In general, rumen bacteria grew with similar efficiency on cellulose and glucose, regardless of how efficiency is measured. Although glucose supported a slightly larger bacterial mass, this advantage was offset by more carbohydrate being fermented. When data were fit to a mathematical model, neither the maintenance coefficient nor the maximum growth efficiency differed between substrates. These parameters had large SE, but it still suggests that source of carbohydrate is not

a major factor determining the efficiency of microbial growth in mixed communities.

Fermentation Explorer Predicts Traits of Microbes in Fermentors

Our laboratory recently developed Fermentation Explorer, an online resource for predicting traits of fermentative microbes. We applied this resource to the current study.

Using names (taxonomy) as inputs, Fermentation Explorer predicted 18 traits of microbes in glucose- and cellulose-fed fermentors (Figure 5A and Supplemental Dataset S2). Many of these predictions matched findings from our experiments. In both glucose- and cellulose-fed fermentors, most microbes were predicted to carry out fermentation, with a small fraction predicted to carry out methanogenesis. Microbes in glucose-fed fermentors were predicted to use glucose (or carbohydrates), whereas microbes in cellulose-fed fermentors were predicted to also use cellulose and cellobiose. It predicted some end products observed in fermentors (e.g., acetate and propionate), though some end products were predicted but not observed (and vice versa). Most microbes were predicted to be anaerobic and isolated from cattle, the rumen, or other gastrointestinal sites.

Fermentation Explorer makes predictions for individual taxa, pinpointing the role of each in fermentation (Figure 5B). For cellulose-fed fermentors, for example, it revealed *Fibrobacter* fermented cellulose in part to succinate, which was converted to propionate by *Succiniclasicum*. For glucose-fed fermentors, the role of some microbes is clear, but it is unclear which microbes form butyrate and caproate.

In sum, the Fermentation Explorer-predicted traits of microbes were largely consistent with experimental observations. It accurately predicted substrates used by microbes in glucose- and cellulose-fed fermentors. It also revealed the contributions of individual taxa, including *Fibrobacter* and *Succiniclasicum*. It left producers of butyrate and caproate unresolved, helping to focus future work on identifying these microbes.

DISCUSSION

Microbes growing in the rumen account for the majority of protein absorbed by the ruminant. Improving efficiency of microbial growth could increase protein available to the ruminant in turn, benefiting the livestock industry. However, factors affecting this efficiency are still not fully understood. One particular gap in our knowledge is the influence of type of carbohydrate. Nearly all studies have grown bacteria on simple sugars (Russell and Wallace, 1997), even though ruminant diets are high in cellulose and other fiber. The few studies that have used

cellulose have used pure cultures of bacteria (Weimer et al., 1991; Shi and Weimer, 1992; Weimer, 1993), and they did not include simple sugars for comparison.

One barrier to studying this topic is the difficulty in feeding cellulose to fermentors. Although sugars can be readily fed as a solution, cellulose is insoluble and settles quickly in water. We handled this issue by building a device that delivers a slurry of cellulose with a syringe. It is simpler than devices used previously, such as one that delivered cellulose through tubes in segmented droplets of liquid (Weimer et al., 1991). Because of this simplicity, we were able to scale it up to multiple fermentors. One limitation of our system is that it could maintain a well-mixed slurry with only fine particles of cellulose; more work is needed to develop a device capable of feeding larger particles more typical of the rumen environment.

With this feeding device in place, we operated fermentors similarly to Isaacson et al. (1975). The study of Isaacson et al. (1975) measured efficiency of mixed rumen bacteria, although it fed glucose alone. Our study used the same rumen fluid-based buffer and the same range of dilution rates as in Isaacson et al. (1975). Further, our target feeding rate (30 mmol of hexose $L^{-1} \cdot d^{-1}$) fell within the range of this previous study (2.8 to 72 mmol of hexose $L^{-1} \cdot d^{-1}$). The similarity of our systems not only supports our approach, but also allows us to directly compare results.

We found that our system supported distinct communities of glucose- and cellulose-fermenting bacteria. In glucose-fed fermentors, the community contained only a few types of bacteria, as in Isaacson et al. (1975). These bacteria belonged to principally to *Streptococcus* and *Kandleria*, which ferment glucose (Sharpe et al., 1973; Russell and Robinson, 1984). The community in cellulose-fed fermentors, by contrast, was much more complex. It included *Fibrobacter* and *Ruminococcus*, which ferment cellulose (Bryant and Doetsch, 1954; Bryant et al., 1958). Additionally, it included *Aristaeella*, which ferment cellobiose (Mahoney-Kurpe et al., 2023). These fermentors also included unclassified groups, indicating that a rich community had developed.

Each community of bacteria carried out a vigorous fermentation of their respective substrate. Glucose was completely fermented, as in the experiment of Isaacson et al. (1975), and cellulose was digested quickly (24.1% h^{-1}). The particle size of cellulose, which was smaller than in the rumen, likely enabled rapid digestion. Further, digestion was faster than observed with pure cultures of bacteria fermentors, which were provided the same particle size of cellulose as in our study (Weimer et al., 1991; Shi and Weimer, 1992; Weimer, 1993). In this case, our faster digestion rates may have resulted from our system involving a mixed community or having uncharacterized organisms capable of faster degrada-

tion. Regardless of the reason, the fast digestion allowed cellulose-degrading organisms to be maintained at dilution rates higher than previously achieved (Weimer et al., 1991; Shi and Weimer, 1992; Weimer, 1993).

The roles of specific organisms in fermentation were pinpointed by Fermentation Explorer (Hackmann and Zhang, 2023). For cellulose-fed fermentors, a relatively full picture of fermentation emerged. For glucose-fed fermentors, the picture was less complete, and it was unclear which organisms form butyrate and caproate. Although *Megasphaera* can form caproate (Marounek et al., 1989; Jeon et al., 2017), they were in low abundance in our fermentors, and they were frequently absent from those that produced caproate. Caproate is not usually detected in the rumen (Foroutan et al., 2020) and is a potential biofuel (Harvey and Meylemans, 2014). Thus, it is important for future work to identify which organisms in our study form it and how. We hypothesize butyrate and caproate are formed from lactate, which is biochemically possible (Wang et al., 2022) and would partly explain why lactate did not reach high concentrations.

With our fermentor system in place, we were able to compare efficiency of bacterial growth on glucose versus cellulose. Overall, we found efficiency did not differ by source of carbohydrate. Across dilution rates, values of efficiency ranged from 27.8 to 63.8 g of dry cells (mol hexose fermented) $^{-1}$ for glucose and from 27.2 to 79.5 g of dry cells (mol hexose fermented) $^{-1}$ for cellulose. This generally falls within the range observed in Isaacson et al. (1975; 42.2 to 83.9 g of dry cells [mol hexose fermented] $^{-1}$). This shows that cells in our experiment grew similarly to those in a previous study using glucose alone.

To analyze efficiency in more detail, we use the model of Pirt (1965) to estimate values of the maintenance coefficient and maximum efficiency. These parameters did not differ by source of carbohydrate, though SE were large. Our values of maintenance were higher than in Isaacson et al. (1975; 0.26 mmol hexose [g of dry cells] $^{-1} h^{-1}$), which fed glucose to mixed bacteria. They were more similar to studies involving pure cultures of bacteria (Supplemental Table S2), though the range across those studies is large. Our values of maximum growth efficiency were similar to Isaacson et al. (1975; 89.3 g of dry cells [mol hexose fermented] $^{-1}$). This analysis indicates that efficiency was similar across carbohydrate sources, with parameters for efficiency being reasonably similar to previous studies.

Several factors affect efficiency of growth, and from those factors alone, it is hard to predict if cells would grow more efficiently on cellulose or glucose. One factor favoring more efficient growth on cellulose is that cellulose fermentation may release more ATP. When cellulose is hydrolyzed to cellobiose and to glucose, ATP is released by cellobiose phosphorylase (see Supplemental

Table S1). No equivalent ATP is released during glucose fermentation. One factor favoring less efficient growth on cellulose is that synthesis of biomass likely requires more ATP. Cellulolytic bacteria synthesize large amounts of extracellular material (Weimer et al., 2006), and this has an energetic cost. Our study shows that these factors must balance out, resulting in similar efficiency of growth on cellulose and glucose.

By providing cellulose, our experiment is meant to mimic conditions in the rumen more closely than previous ones. However, the conditions in our fermentors are still artificial. Because of the fine particle size of cellulose, digestibility of cellulose in our experiment generally falls above the digestibility of NDF in the rumen (27.0%–80.9%; Huhtanen et al., 2010). Additionally, dilution rates for cellulose-fed fermentors go above values observed in the rumen for forage particles (1.5%–7.8%·h⁻¹; Seo et al., 2006). Likewise, dilution rates for glucose-fed fermentors go below those for rumen liquids (2.6%–22.4%·h⁻¹; Seo et al., 2006). Other artificial conditions of fermentors, such as low feeding rates (Hristov et al., 2012), also apply to our experiment.

Under the conditions in our experiment, efficiency of bacterial growth did not differ by source of carbohydrate. Our conclusions need to be qualified by the relatively large SE in measurement. With these qualifications, our study suggests factors other than carbohydrate may be more important in determining efficiency of microbial growth in the rumen. It underscores that source of carbohydrate still has a strong effect on the composition of the microbial community and its fermentation products.

NOTES

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Nonstandard abbreviations used: ASV = amplicon sequence variant; Ce = cellulose; Glc = glucose; LPSN = List of Prokaryotic names with Standing in

Nomenclature; max = maximum; NMDS = nonmetric multidimensional scaling.

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